刘波

181-7531-0983 | bo.liu@tum.de | Google Scholar | 光通信均衡器 | AI 硬件加速 | IC 设计

教育背景

慕尼黑工业大学 | 计算机工程 | 工学博士 2021.06 - 2026.03

方向：高效神经网络加速器、高效光通信均衡器（与华为慕尼黑研究中心合作）

成果：论文 6 篇（一作 3 篇），专利 1 项；IEEE TCASAI 创刊论文、OFC/ECOC 光通信顶会论文；代码开源

东南大学 | 微电子学 | 工学硕士 2017.09 - 2020.06

方向：语音处理算法的硬件设计与实现

成果：论文 2 篇（一作 1 篇），专利 1 项；FPGA 创新设计邀请赛全国一等奖；CyberC 最佳学生论文奖；一等奖学金

吉林大学 | 微电子学 | 理学学士 2012.09 - 2016.06

GPA: 3.45/4.0 (24/137，前 20%)；**获奖**：校一/二等奖学金，校/院优秀学生

项目经历

面向 800 Gb/s PAM4 CWDM4 的神经网络光通信均衡器设计 | 博士课题 2021.06 - 2022.12

- 针对不同波长链路的**均衡器重复设计**问题，提出基于**多任务学习**的**单均衡器多波长共享架构**
- 针对**光/电信号吞吐差异**带来的**硬件并行**需求，提出基于**多输出学习**的**多输出并行均衡器架构**
- 在 **45 nm**、**1 GHz** 下，与传统方法性能持平，**面积降低 77%，功耗降低 61%**，仅配置 bias 即可适配不同波长

高效 MAC Array 设计和高效神经网络加速器 | 博士课题 2023.01 - 2024.11

- 面向神经网络硬件部署中乘加单元面积大、功耗高的问题，设计 **广义有权编码 MAC Array**
- 在 **15 nm**、**5 GHz**、**64×64 MAC Array** 实现中，相比传统 Systolic Array，**功耗降低 43.10%，面积降低 23.69%**

低信噪比下的高精度语音活动性检测的 VLSI 设计与实现 | 硕士毕设 2018.11 - 2020.05

- 面向高背景噪声场景下语音活动检测准确率低、硬件代价高的问题，设计低功耗 VLSI 实现方案
- 相比现有方案，检测准确率提升至 **85%–94%，功耗降低 30%，面积减少 26%**

基于 LSTM 的语音关键词识别的 FPGA 设计 | 硕士竞赛 2018.09 - 2018.12

- 训练 LSTM，完成 Verilog 设计及验证，部署到 Artix-7 开发板，获 FPGA 创新设计邀请赛**全国一等奖**

论文

共发表论文 8 篇，其中一作/二作覆盖 IEEE TCASAI、OFC、ECOC、JLT、OECC、CyberC 等，部分论文如下

[1] **Bo Liu**, et al., "Area-Efficient Neural Network CD Equalizer for 4×200 Gb/s PAM4 CWDM4 Systems," in *Optical Fiber Communication Conference (OFC)*, 2023. **(光通信顶会，获欧洲专利)**

[2] **Bo Liu**, et al., "Area-Efficient Hardware Parallelization of Neural Network CD Equalizers for 4×200 Gb/s PAM4 CWDM4 Systems," in *European Conference on Optical Communication (ECOC)*, 2023. **(光通信顶会)**

[3] Christian Blümm, **Bo Liu**, et al., "800 Gb/s PAM4 Transmission Over 10 km SSMF Enabled by Low-Complex Duobinary Neural Network Equalization," in *European Conference on Optical Communication (ECOC)*, 2022.

[4] Christian Blümm, **Bo Liu**, et al., "Hardware-Efficient Duobinary Neural Network Equalizers for 800 Gb/s IM/DD PAM4 Transmission over 10 km SSMF," *Journal of Lightwave Technology (JLT)*, 2023.

[5] Christian Blümm, **Bo Liu**, et al., "Towards Neural Network Equalizer Implementations for IM/DD Transceivers," in *OptoElectronics and Communications Conference (OECC)*, 2023.

专利

[1] **Bo Liu**, et al., "Device and Method for Performing Communication Channel Equalization in a Plurality of Channel Scenarios Using a Neural Network-Based Equalizer," ***WO2024170050A1***, 2024. **(欧洲专利)**

Bo Liu

(+86)181-7531-0983 | bo.liu@tum.de | Google Scholar | Optical Equalizer | AI Hardware

Education

Technical University of Munich | Computer Engineering | *Dr.-Ing.* 2021.06 - 2026.03

Research: Efficient DNN acceleration and optical fiber communication equalization (work with Huawei)

Results: 6 papers (3 first-authored), 1 patent; IEEE TCASAI/OFC/ECOC/JLT/OECC; Open-source code

Southeast University | Microelectronics | *M.-Eng.* 2017.09 - 2020.06

Research: Chip design and implementation for speech processing algorithms

Results: 2 papers, 1 patent; National FPGA Innovation Contest First Prize; Best Student Paper; 1st Scholarship

Jilin University | Microelectronics | *B.-Sc.* 2012.09 - 2016.06

GPA: 3.45/4.0 (24/137, top 20%); **Awards**: 1st and 2nd Scholarships; Outstanding Student Honor

Projects

Neural Network Optical Equalizer for 800 Gb/s PAM4 CWDM4 | *Ph.D. Research* 2021.06 - 2022.12

- For redundant per-channel designs, proposed multi-task-learning-based shared equalizer across wavelengths.
- For high-throughput hardware parallelization, proposed multi-output-learning-based parallel equalizer architecture.
- Achieved **77% area reduction** and **61% power reduction** at **45 nm, 1 GHz** with maintained performance.

Efficient MAC Array Design for Neural Network Acceleration | *Ph.D. Research* 2023.01 - 2024.11

- Designed a **general-weighted-encoding MAC array** to reduce the area and power cost of MAC units.
- Achieved **43.10% lower power** and **23.69% smaller area** in a **15 nm, 5 GHz, 64×64** implementation.

Low-Power VLSI Design for VAD under Low SNR | *Master's Thesis* 2018.11 - 2020.05

- Designed a low-power VLSI for VAD in high-noise environments, addressing low accuracy and high hardware cost.
- Achieved **85%–94%** detection accuracy, with **30% lower power** and **26% smaller area** than prior designs.

FPGA Design for LSTM-Based Keyword Spotting | *Master's Competition Project* 2018.09 - 2018.12

- Trained LSTM model, implemented in Verilog, deployed on Artix-7. Won the **First Prize** at FPGA Contest.

Papers

Published 8 papers, with first-/second-authored at/in IEEE TCASAI, OFC, ECOC, JLT, OECC, CyberC. Selected:

[1] **Bo Liu**, et al., "Area-Efficient Neural Network CD Equalizer for 4×200 Gb/s PAM4 CWDM4 Systems," in *Optical Fiber Communication Conference (OFC)*, 2023. **(Top conference, European patent)**

[2] **Bo Liu**, et al., "Area-Efficient Hardware Parallelization of Neural Network CD Equalizers for 4×200 Gb/s PAM4 CWDM4 Systems," in *European Conference on Optical Communication (ECOC)*, 2023. **(Top conference)**

[3] Christian Blümm, **Bo Liu**, et al., "800 Gb/s PAM4 Transmission Over 10 km SSMF Enabled by Low-Complex Duobinary Neural Network Equalization," in *European Conference on Optical Communication (ECOC)*, 2022.

[4] Christian Blümm, **Bo Liu**, et al., "Hardware-Efficient Duobinary Neural Network Equalizers for 800 Gb/s IM/DD PAM4 Transmission over 10 km SSMF," *Journal of Lightwave Technology (JLT)*, 2023.

[5] Christian Blümm, **Bo Liu**, et al., "Towards Neural Network Equalizer Implementations for IM/DD Transceivers," in *OptoElectronics and Communications Conference (OECC)*, 2023.

Patent

[1] **Bo Liu**, et al., "Device and Method for Performing Communication Channel Equalization in a Plurality of Channel Scenarios Using a Neural Network-Based Equalizer," ***WO2024170050A1***, 2024. **(Europe)**